MIT 8103 Advanced Database Systems

Library Management System · Ibukunoluwa Oluwashola · Student ID 301926692

Case study

My case study is Library Management System. I designed the database for library staff managing checkout and return and library managers reviewing circulation, with separate tables named member, book and loan. It supports these operations: record a loan linking one member to one book; record a return by adding its return date; report loans by member or book; separately benchmark the loan-date range filter against 50000 synthetic loan rows, comparing plans with and without a B-tree index on loan_date.

Portfolio 1: Database Design and Modelling

I organised the relational design into 3 tables. Primary keys identify records, foreign keys preserve the links, and checks enforce the stated business rules. A book may have at most one loan whose returned_on value is null. The normalisation examples show the sample data at 1NF, 2NF and 3NF. All 10 deliberately invalid inserts were rejected without changing the valid sample rows.

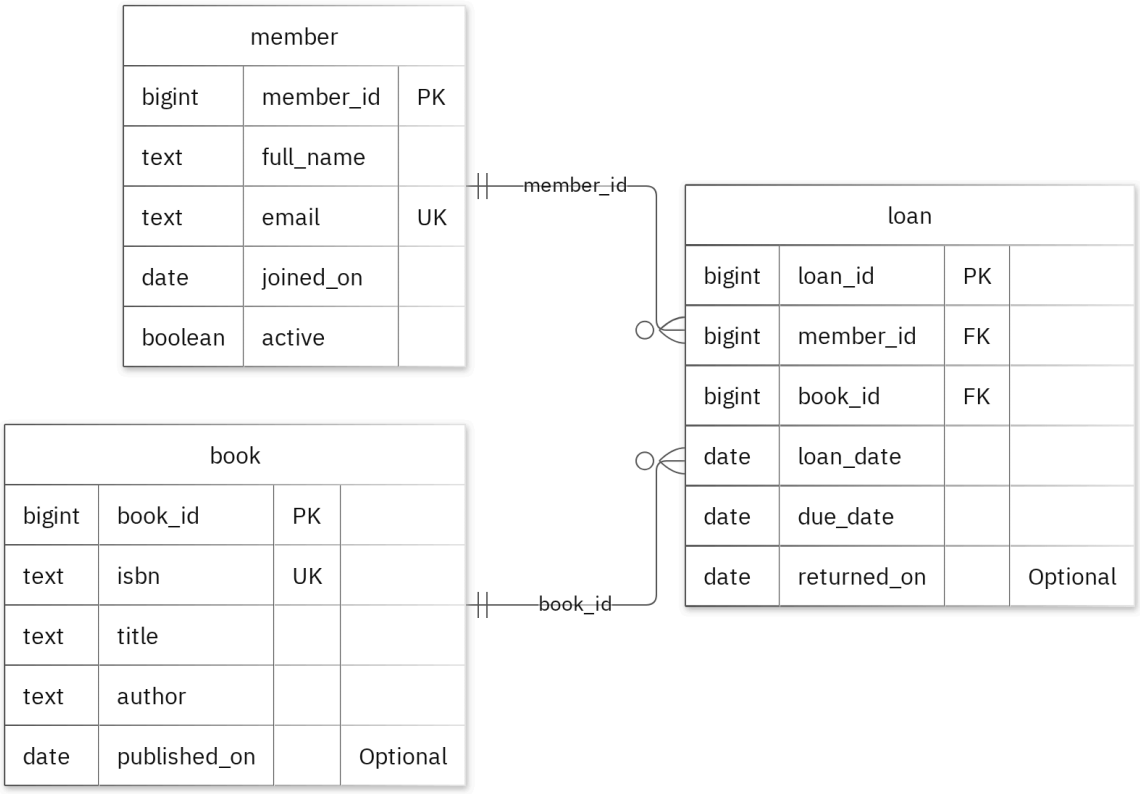


Figure 1. Tables, keys and relationships. PK: primary key; FK: foreign key; UK: unique value.

Portfolio 2: Query Processing and Optimisation

I ran three SQL queries against the sample records. The join returned 2 rows and included Avery Chen. For the index comparison, I used 50,000 separate performance rows. The median lookup time changed from 2.26 ms to 0.276 ms. Both phases returned the same records. The measured plan used the new index. These timings apply to this lookup and environment.

Portfolio 3: Transactions and Concurrency

I used a temporary counter on loan to test COMMIT, ROLLBACK and READ COMMITTED isolation. The uncommitted update remained hidden from the other session, and ROLLBACK restored the earlier committed value. Without locking, two increments left the counter at 2. With SELECT FOR UPDATE, Session B waited for Session A and the final value was 3. The helper is confined to this exercise; the case's original business fields stay unchanged.

Portfolio 4: NoSQL and Advanced Data Models

I used loan_events for the case's event documents. MongoDB inserted 4 sample events and returned the expected results for 4 queries. Validation rejected an invalid date type and a duplicate event ID. Embedded snapshots keep related details with an event, while the details field can differ by event type. These are fictional events; PostgreSQL continues to enforce its relational rules.

Repository and evidence

https://github.com/Betty-Ibk/Oluwashola_Ibukunoluwa_301926692

The portfolio folders contain the SQL and JavaScript, explanations, query plans, result tables and screenshots.

Portfolio 5: Distributed and Cloud Database Exercise

I proposed three MongoDB replica members across separate failure zones. The discussion covers primary elections, majority acknowledgement, secondary lag and sharding trade-offs for loan_events. Replication supports availability, while backups address recovery. PostgreSQL needs its own availability plan. This is the permitted discussion option; no live replica set or failover test was deployed.

Individual reflection

I kept member, book, loan in separate tables so their details could be maintained without repeating them in every related record. A book may have at most one loan whose returned_on value is null. The constraint tests check this alongside the required values and record links. One issue illustrated by the two-session exercise is that both sessions can read the same value before either update is committed. Locking the row before reading preserves both increments. The index comparison also shows why I need to check the execution plan as well as the final answer: the same result can require different amounts of database work.

AI-use declaration

I used GitHub Copilot and the portfolio generator to help prepare the database model, scripts and explanations. The generator ran the PostgreSQL and MongoDB exercises, checked their outputs and captured the result images. I remain responsible for checking and explaining the submitted work.

References

- PostgreSQL: constraints
- PostgreSQL: EXPLAIN
- PostgreSQL: isolation
- PostgreSQL: row locks
- MongoDB: data modelling
- MongoDB: replication
- MongoDB: sharding
- Mermaid: ER diagrams